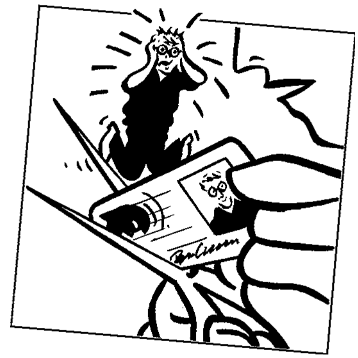
CHAPTER 23
credit risk



The aim of this Chapter...

...is to introduce the concept of creditworthiness and default into our financial modeling. We will also be seeing a new random process, the Poisson process, which is useful for modeling sudden, unexpected changes of state.

In this Chapter...

- the Merton model
- models for instantaneous and exogenous risk of default
- stochastic risk of default and implied risk of default
- credit ratings
- how to model change of rating

23.1 INTRODUCTION

In this chapter I describe some ways of looking at default. A recent approach to this subject is to model default as a completely exogenous event, i.e. a bit like the tossing of a coin or the appearance of zero on a roulette wheel, and having nothing to do with how well the company or country is doing. Typically, one then infers from risky bond prices the probability of default as perceived by the market. I'm not wild about this idea but it is very popular.[1]

Later in this chapter I describe the rating service provided by Standard & Poor's and Moody's, for example. These ratings provide a published estimate of the relative creditworthiness of firms.

But first we look at a classic option approach applied to debt valuation, the Merton model.

23.2 THE MERTON MODEL: EQUITY AS AN OPTION ON A COMPANY'S ASSETS

The Merton model shows very elegantly how the equity of a company can be thought of as a simple call option on the assets of the company. He starts off by assuming that the assets of the company A follow a random walk

$$dA = \mu A\, dt + \sigma A\, dX.$$

Clearly, the value of the equity equals assets less liabilities:

$$S = A - V.$$

Here S is the value of the equity (just the share price, I've assumed that there is only one share for simplicity) and V the value of the debt.

At maturity of this debt

$$S(A, T) = \max(A - D, 0) \quad \text{and} \quad V(A, T) = \min(D, A),$$

where D is the amount of the debt, to be paid back at time T.

We can now apply ideas from the Black–Scholes model for options, namely the ideas of delta hedging and no arbitrage. Set up a portfolio consisting of the debt and short a quantity Δ of the equity:

$$\Pi = V - \Delta S.$$

We are going to hedge the debt with the stock. (Don't worry about buying and selling a fraction of the one share, in reality there would be millions of shares issued.)

This portfolio changes by

$$d\Pi = dV - \Delta\, dS.$$

[1] Bankers are paid such obscene amounts of money, you'd hope they'd feel obliged to think originally once in a while, wouldn't you? Apparently, this is not the case.

So applying Itô's lemma

$$d\Pi = \left(\frac{\partial V}{\partial t} + \tfrac{1}{2}\sigma^2 A^2 \frac{\partial^2 V}{\partial A^2} - \Delta \frac{\partial S}{\partial t} - \tfrac{1}{2}\Delta\sigma^2 A^2 \frac{\partial^2 S}{\partial A^2} \right) dt + \left(\frac{\partial V}{\partial A} - \Delta \frac{\partial S}{\partial A} \right) dA.$$

Now choose

$$\Delta = \frac{\dfrac{\partial V}{\partial A}}{\dfrac{\partial S}{\partial A}}$$

to eliminate risk, and set the return $d\Pi$ equal to the risk-free return $r\Pi\, dt \ldots$

The end result is, for the current value of the debt V

$$\frac{\partial V}{\partial t} + \tfrac{1}{2}\sigma^2 A^2 \frac{\partial^2 V}{\partial A^2} + rA \frac{\partial V}{\partial A} - rA = 0$$

subject to

$$V(A, T) = \min(D, A)$$

and exactly the same partial differential equation for the equity of the firm S but with

$$S(A, T) = \max(A - D, 0).$$

Recognize this? The problem for S is exactly that for a call option, but now we have S instead of the option value, the underlying variable are the assets A and the strike is D, the debt. The solution for the equity value is precisely that for a call option.

And why is this relevant for credit risk? If this 'call option on the assets' ends up out of the money then it is equivalent to a default. So from this model we can find the (risk-neutral) probability of this happening, and if it does happen, what is the recovery rate.

23.3 RISKY BONDS

If you are a company wanting to expand, but without the necessary capital, you could borrow the capital, intending to pay it back with interest at some time in the future. There is a chance, however, that before you pay off the debt the company may have got into financial difficulties or even gone bankrupt. If this happens, the debt may never be paid off. Because of this risk of default, the lender will require a significantly higher interest rate than if he were lending to a very reliable borrower such as the US government.

The real situation is, of course, more complicated than this. In practice it is not just a case of all or nothing. This brings us to the ideas of the seniority of debt and the partial payment of debt.

Firms typically issue bonds with a legal ranking, determining which of the bonds take priority in the event of bankruptcy or inability to repay. The higher the priority, the more

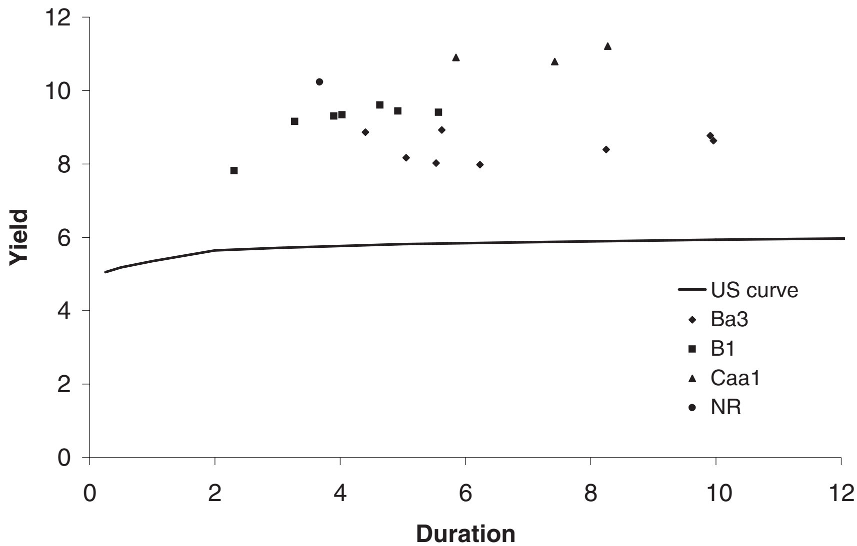


Figure 23.1 Yield versus duration for some risky bonds.

likely the debt is to be repaid, the higher the bond value and the lower its yield. In the event of bankruptcy there is typically a long, drawn out battle over which creditors get paid. It is usual, even after many years, for creditors to get *some* of their money back. Then the question is how much and how soon? It is also possible for the repayment to be another risky bond, this would correspond to a refinancing of the debt. For example, the debt could not be paid off at the planned time so instead a new bond gets issued entitling the holder to an amount further in the future.

In Figure 23.1 is shown the yield versus duration, calculated by the methods of Chapter 14, for some risky bonds. In this figure the bonds have been ranked according to their estimated riskiness. We will discuss this later, for the moment you just need to know that Ba3 is considered to be less risky than Caa1 and this is reflected in its smaller yield spread over the risk-free curve.

The problem that we examine in this chapter is the modeling of the risk of default and thus the fair value of risky bonds. Conversely, if we know the value of a bond, does this tell us anything about the likelihood of default?

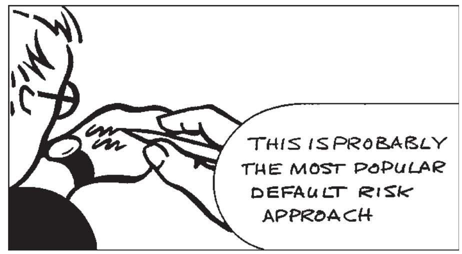
23.4 MODELING THE RISK OF DEFAULT

The models that I now describe assume that the likelihood of default is exogenous. These instantaneous risk of default models are simple to use and are therefore the most popular type of credit risk models. In its simplest form the time at which default occurs is completely exogenous. For example, we could roll a die once a month, and when a 1 is thrown the company defaults. This illustrates the exogeneity of the default and also its randomness, a Poisson process is a typical choice for the time of default. We will see that when the intensity of the Poisson process is constant (as in the die example), the pricing of risky bonds amounts to the addition of a time-independent spread to the bond yield. We will also see models for which the intensity is itself a random variable.

A refinement of the modeling that we also consider is the regrading of bonds. There are agents, such as Standard & Poor's and Moody's, who classify bonds according to their estimate of their risk of default. A bond may start life with a high rating, for example, but may find itself regraded due to the performance of the issuing firm. Such a regrading will have a major effect on the perceived risk of default and on the price of the bond. I will describe a simple model for the rerating of risky bonds.

At the end of the chapter we take a quick look at modeling the interrelationships between many risky instruments via copulas.

23.5 THE POISSON PROCESS AND THE INSTANTANEOUS RISK OF DEFAULT



One approach to the modeling of credit risk is via the **instantaneous risk of default**, p. If at time t the company has not defaulted and the instantaneous risk of default is p then the probability of default between times t and $t + dt$ is $p\,dt$. This is an example of a Poisson process; nothing happens for a while, then there is a *sudden* change of state. This is a continuous-time version of our earlier model of throwing a die.

Time Out...

Poisson processes

The basic building block for the random walks we have considered so far is continuous Brownian motion based on the Normally distributed increment. Another stochastic process that is useful in finance is the Poisson process.

A Poisson process dq is defined by

$$dq = \begin{cases} 0 & \text{with probability } 1 - \lambda\, dt \\ 1 & \text{with probability } \lambda\, dt. \end{cases}$$

There is therefore a probability $\lambda\, dt$ of a jump in q in the time step dt. The parameter λ is called the **intensity** of the Poisson process. The scaling of the probability of a jump with the size of the time step is important in making the resulting process 'sensible,' i.e. there being a finite chance of a jump occurring in a finite time, with q not becoming infinite.

Spreadsheet simulation

The simplest example to start with is to take p constant. In this case we can easily determine the risk of default before time T. We do this as follows.

Let $P(t;T)$ be the probability that the company does not default before time T given that it has not defaulted at time t. The probability of default between later times t' and $t'+dt'$ is the product of $p\,dt$ and the probability that the company has not defaulted up until time t'. Thus,

$$P(t'+dt',T)-P(t',T)=p\;dt\;P(t',T).$$

Expanding this for a small time step results in the ordinary differential equation representing the rate of change of the required probability:

$$\frac{\partial P}{\partial t'}=pP(t';T).$$

If the company starts out not in default then $P(T;T)=1$. The solution of this problem is

$$e^{-p(T-t)}.$$

The value of a zero-coupon bond paying \$1 at time T could therefore be modeled by taking the present value of the *expected* cashflow. This results in a value of

$$e^{-p(T-t)}Z, \tag{23.1}$$

where Z is the value of a riskless zero-coupon bond of the same maturity as the risky bond. Note that this does not put any value on the risk taken. The yield to maturity on this bond is now given by

$$-\frac{\log(e^{-p(T-t)}Z)}{T-t}=-\frac{\log Z}{T-t}+p.$$

Thus the effect of the risk of default on the yield is to add a spread of p. In this simple model, the spread will be constant across all maturities.

Now we apply this to derivatives, including risky bonds. We will assume that the spot interest rate is stochastic. For simplicity we will assume that there is no correlation between the diffusive change in the spot interest rate and the Poisson process.

Construct a 'hedged' portfolio:

$$\Pi=V(r,p,t)-\Delta Z(r,t).$$

Consider how this changes in a time step. See Figure 23.2 for a diagram illustrating the analysis below.

There is a probability of $(1-p\,dt)$ that the bond does not default. In this case the change in the value of the portfolio during a time step is

$$d\Pi=\left(\frac{\partial V}{\partial t}+\tfrac{1}{2}w^2\frac{\partial^2 V}{\partial r^2}\right)dt+\frac{\partial V}{\partial r}dr+-\Delta\left(\left(\frac{\partial Z}{\partial t}+\tfrac{1}{2}w^2\frac{\partial^2 Z}{\partial r^2}\right)dt+\frac{\partial Z}{\partial r}dr\right). \tag{23.2}$$

Choose Δ to eliminate the risky dr term.

On the other hand, if the bond defaults, with a probability of $p\,dt$, then the change in the value of the portfolio is

$$d\Pi=-V+O(dt^{1/2}). \tag{23.3}$$

This is due to the sudden loss of the risky bond, the other terms are small in comparison.

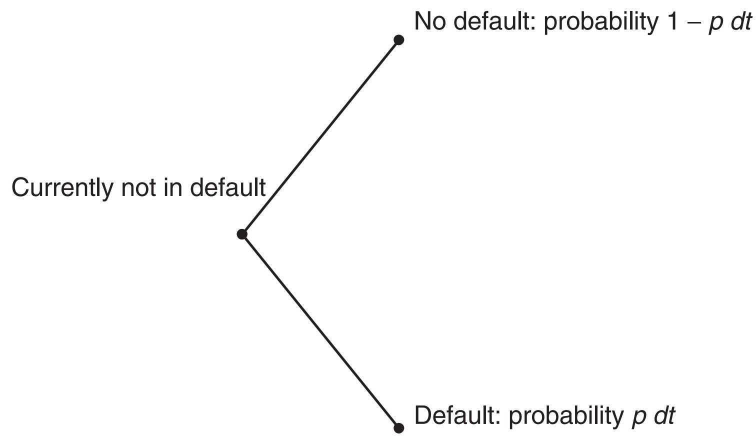


Figure 23.2 A schematic diagram showing the two possible situations: default and no default.

Taking expectations and using the bond pricing equation for the riskless bond, we find that the value of the risky bond satisfies

$$\boxed{\frac{\partial V}{\partial t} + \frac{1}{2}w^2\frac{\partial^2 V}{\partial r^2} + (u - \lambda w)\frac{\partial V}{\partial r} - (r + p)V = 0.} \qquad (23.4)$$

Observe that the spread has been added to the discounting term.

Time Out...

$\boldsymbol{r \to r + p}$

The risk of default has been added to the discounting term. Thus all cashflows are going to be discounted at a rate of $r + p$ instead of r. We can interpret p as a 'spread.' With p being positive this means that all positive cashflows have less present value than if there were no credit risk. The simplicity of this model is the reason for its popularity. (It is not popular because it is a good model.)

Example A four-year US government risk-free zero-coupon bond has a principal of \$100 and a current value of \$81.11. A similar bond issued by a less reliable source has a value of \$77.72. What do these numbers tell you? The yield to maturity of the US government bond is 5.2%. The yield to maturity of the risky bond is 6.3%. The spread of 1.1% between these two means that the market expects (ish) that there is a $1 - e^{-0.011} \approx 1.11\%$ chance per year of the company defaulting.

How would you change this model so that negative cashflows are not affected? After all, why would a company refuse a payment in its favor?

The portfolio is only hedged against moves in the interest rate (assuming that the market price of risk is known, big assumption) and not the event of default. I'll come back to this point below.

This model is the most basic for the instantaneous risk of default. It gives a very simple relationship between a risk-free and a risky bond. There is only one new parameter to estimate, p.

To see whether this is a realistic model for the expectations of the market we take a quick look at the valuation of Brady bonds. In particular we examine the market price of Latin American par bonds, described in full later in this chapter. For the moment, we just need to know that these bonds have interest payments and final return of principal denominated in US dollars. If the above is a good model of market expectations with constant p then we would find a very simple relationship between interest rates in the US and the value of Brady bonds. To find the Brady bond value perform the following:

1. Find the risk-free yield for the maturity of each cashflow in the risky bond.
2. Add a constant spread, p, to each of these yields.
3. Use this new yield to calculate the present value of each cashflow.
4. Sum all the present values.

Conversely, the same procedure can be used to determine the value of p from the market price of the Brady bond: this would be the **implied risk of default**. In Figure 23.3 are shown the implied risks of default for the par bonds of Argentina, Brazil, Mexico and Venezuela using the above procedure and assuming a constant p.

In this simple model we have assumed that the instantaneous risk of default is constant (different for each country) through time. However, from Figure 23.3 we can see that, if we believe the market prices of the Brady bonds, this assumption is incorrect: the market prices are inconsistent with a constant p. This will be our motivation for the stochastic risk of default model which we will see in a moment. Nevertheless, supposing that the figure represents, in some sense, the views of the market (and this constant p model *is* used in practice) we draw a few conclusions from this figure before moving on.

The first point to notice in the graph is the perceived risk of Venezuela, which is consistently greater than the three other countries. Venezuela's risk peaked in July 1994, nine months before the rest of South America, but this had absolutely no effect on the other countries.

The next, and most important, thing to notice is the 'tequila effect' in all the Latin markets. The tequila crisis began with a 50% devaluation of the Mexican peso in December 1994.

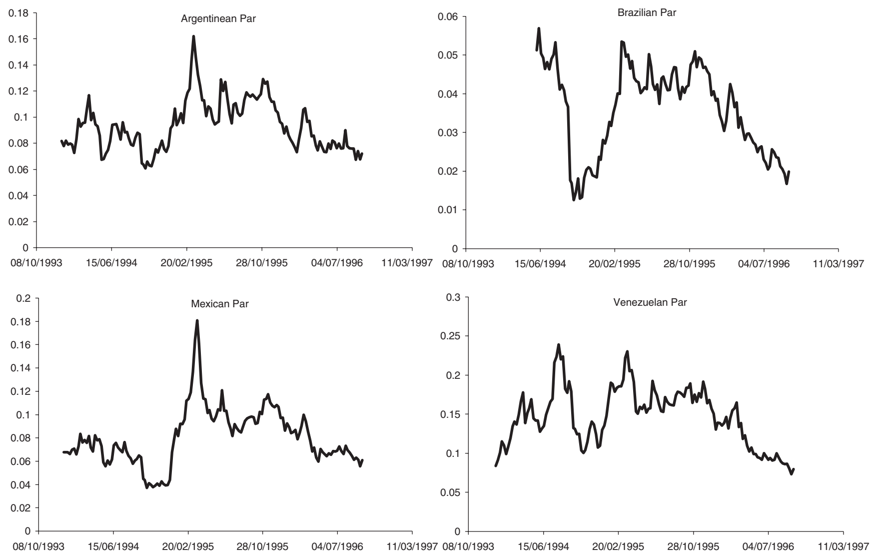
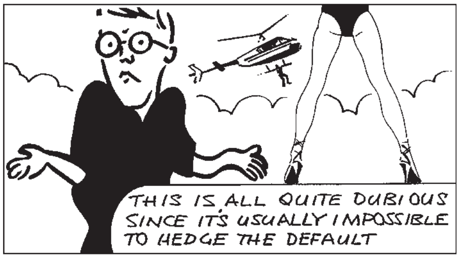


Figure 23.3 The implied risk of default for the par bonds of Argentina, Brazil, Mexico and Venezuela assuming constant p.

Markets followed suit by plunging. Before December 1994 we can see a constant spread between Mexico and Argentina and a contracting spread between Brazil and Argentina. The consequences of tequila were felt through all the first quarter of 1995 and had a knock-on effect throughout South America. In April 1995 the default risks peaked in all the countries apart from Venezuela, but by late 1996 the default risk had almost returned to pre-tequila levels in all four countries. By this time, Venezuela's risk had fallen to the same order as the other countries.

23.5.1 A note on hedging



In the above we have not hedged the event of default. This can sometimes be done (kinda), provided we can hedge with other bonds that will default at exactly the same time, and later we'll see how this introduces a market price of default risk term, as might be expected. Usually, though, there are so few risky bonds that hedging default is not possible. For the moment I'm going to work in terms of real expectations on the understanding that in the world of default risk, perfect hedging is rarely possible.

23.6 **TIME-DEPENDENT INTENSITY AND THE TERM STRUCTURE OF DEFAULT**

Suppose that a company issues risky bonds of different maturities. We can deduce from the market's prices of these bonds how the risk of default is perceived to depend on time.

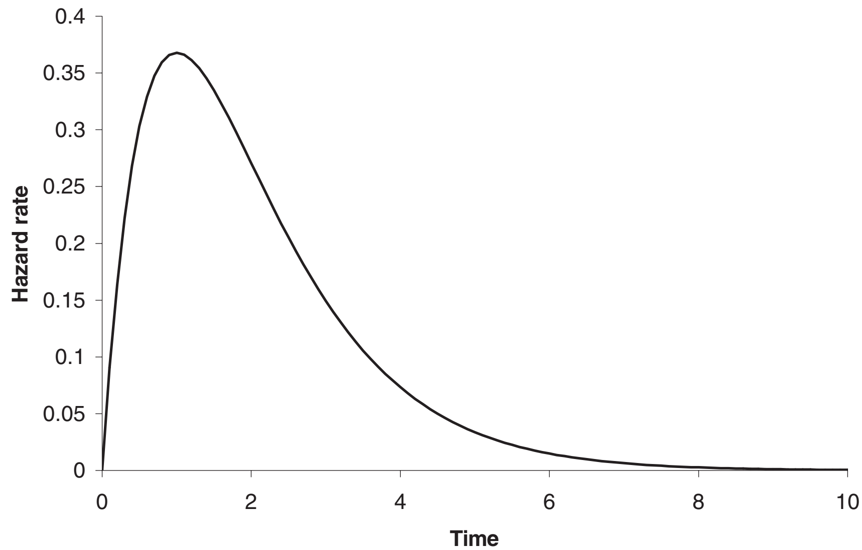


Figure 23.4 A plausible structure for a time-dependent hazard rate.

To make things as simple as possible let's assume that the company issues zero-coupon bonds and that in the event of default in one bond, all other outstanding bonds also default with no recovery rate.

If the risk of default is time dependent, $p(t)$, and uncorrelated with the spot interest rate, then the real expected value of a risky bond paying \$1 at time T is just

$$Ze^{-\int_t^T p(\tau)\,d\tau}.$$

If the market value of the risky bond is Z^* then we can write

$$\int_t^T p(\tau)\,d\tau = \log\left(\frac{Z}{Z^*}\right).$$

Differentiating this with respect to T gives the market's view at the current time t of the **hazard rate** or risk of default at time T. A plausible structure for such a hazard rate is given in Figure 23.4. This figure shows a very small chance of default initially, rising to a maximum before falling off. The company is clearly expected to be around for at least a little while longer, and in the long term it will either have already expired or become very successful. If the area under the curve is finite then there is a finite probability of the company never going bankrupt.

23.7 STOCHASTIC RISK OF DEFAULT

To 'improve' the model, and make it consistent with market prices, we now consider a model in which the instantaneous probability of default is itself random. We assume that it follows a random walk given by

$$dp = \gamma(r, p, t)\,dt + \delta(r, p, t)\,dX_1,$$

with interest rates still given by

$$dr = u(r,t)\,dt + w(r,t)\,dX_2.$$

It is reasonable to expect some interest rate dependence in the risk of default, but not the other way around.

To value our risky zero-coupon bond we construct a portfolio with one of the risky bonds, with value $V(r,p,t)$ (to be determined), and short Δ of a riskless bond, with value $Z(r,t)$ (satisfying our earlier bond pricing equation):

$$\Pi = V(r,p,t) - \Delta Z(r,t).$$

In the next time step either the bond is defaulted or it is not. There is a probability of default of $p\,dt$. We must consider the two cases: default and no default in the next time step. As in the two models above, we take expectations to arrive at an equation for the value of the risky bond.

First, suppose that the bond does not default; this has a probability of $1 - p\,dt$. In this case the change in the value of the portfolio during a time step is

$$d\Pi = \left(\frac{\partial V}{\partial t} + \tfrac{1}{2}w^2\frac{\partial^2 V}{\partial r^2} + \rho w\delta\frac{\partial^2 V}{\partial r\,\partial p} + \tfrac{1}{2}\delta^2\frac{\partial^2 V}{\partial p^2}\right)dt + \frac{\partial V}{\partial r}dr + \frac{\partial V}{\partial p}dp$$
$$- \Delta\left(\left(\frac{\partial Z}{\partial t} + \tfrac{1}{2}w^2\frac{\partial^2 Z}{\partial r^2}\right)dt + \frac{\partial Z}{\partial r}dr\right),$$

where ρ is the correlation between dX_1 and dX_2. Choose Δ to eliminate the risky dr term.

On the other hand, if the bond defaults, with a probability of $p\,dt$, then the change in the value of the portfolio is

$$d\Pi = -V + O(dt^{1/2}).$$

This is due to the sudden loss of the risky bond, the other terms are small in comparison. It is at this point that we could put in a recovery rate, discussed in the next section; as it stands here default means no return whatsoever.

Taking expectations and using the bond pricing equation for the riskless bond, we find that the value of the risky bond satisfies

$$\boxed{\frac{\partial V}{\partial t} + \tfrac{1}{2}w^2\frac{\partial^2 V}{\partial r^2} + \rho w\delta\frac{\partial^2 V}{\partial r\partial p} + \tfrac{1}{2}\delta^2\frac{\partial^2 V}{\partial p^2} + (u - \lambda w)\frac{\partial V}{\partial r} + \gamma\frac{\partial V}{\partial p} - (r+p)V = 0.} \quad (23.5)$$

This equation has final condition

$$V(r,p,T) = 1,$$

if the bond is zero coupon with a principal repayment of \$1.

Equation (23.5) again shows the similarity between the spot interest rate, r, and the hazard rate, p. The equation is remarkably symmetrical in these two variables, the only difference is in the choice of the model for each. In particular, the final term includes a discounting at rate r and also at rate p. These two variables play similar roles in credit risk equations.

Time Out...

Two-factor models

There are three independent variables in this partial differential equation, t, r and p. And one dependent variable V. If we wanted to think of the meaning of such an equation in terms of our usual mountainside, with slopes in the northerly and westerly directions, then we'd have our job cut out. But the principle is the same. Recall our discussion of the explicit finite-difference method on page 145. The idea works even in this higher-dimensional problem.

As a check on this result, return to the simple case of constant p. In the new framework this case is equivalent to $\gamma = \delta = 0$. The solution of (23.5) is easily seen to be

$$e^{-p(T-t)}Z(r,t),$$

as derived earlier.

If γ and δ are independent of r and the correlation coefficient ρ is zero then we can write

$$V(r,p,t) = Z(r,t)H(p,t),$$

where H satisfies

$$\frac{\partial H}{\partial t} + \tfrac{1}{2}\delta^2\frac{\partial^2 H}{\partial p^2} + \gamma\frac{\partial H}{\partial p} - pH = 0,$$

with

$$H(p,T) = 1.$$

In this special, but important case, the default risk decouples from the bond pricing.

23.8 POSITIVE RECOVERY

In default there is usually *some* payment, not all of the money is lost. In Table 23.1, produced by Moody's from historical data, is shown the mean and standard deviations for recovery according to the seniority of the debt. These numbers emphasize the fact that the rate of recovery is itself very uncertain. How can we model a positive recovery?

Table 23.1 Rate of recovery. Source: Moody's.

Class	Mean (%)	Std dev. (%)
Senior secured	53.80	26.86
Senior unsecured	51.13	25.45
Senior subordinated	38.52	23.81
Subordinated	32.74	20.18
Junior subordinated	17.09	10.90

Suppose that on default we know that we will get an amount Q. This will change the partial differential equation. To see this we return to the derivation of Equation (23.4). If there is no default we still have Equation (23.2). However, on default Equation (23.3) becomes instead

$$d\Pi = -V + Q + O(dt^{1/2});$$

we lose the bond but get Q. Taking expectations results in

$$\frac{\partial V}{\partial t} + \tfrac{1}{2}w^2\frac{\partial^2 V}{\partial r^2} + \rho w\delta\frac{\partial^2 V}{\partial r\,\partial p} + \tfrac{1}{2}\delta^2\frac{\partial^2 V}{\partial p^2} + (u - \lambda w)\frac{\partial V}{\partial r} + \gamma\frac{\partial V}{\partial p} - (r + p)V + pQ = 0.$$

Now you are faced with the difficult task of estimating Q, or modeling it as another random variable. It could even be a fraction of V. The last is probably the most sensible approach.

23.9 HEDGING THE DEFAULT

In the above we used riskless bonds to hedge the random movements in the spot interest rate. Can we introduce another risky bond or bonds into the portfolio to help with hedging the default risk? To do this we must assume that default in one bond automatically means default in the other.

Assuming that the risk of default p is constant for simplicity, consider the portfolio

$$\Pi = V - \Delta Z - \Delta_1 V_1,$$

where both V and V_1 are risky.

The choices

$$\Delta_1 = \frac{V}{V_1} \quad \text{and} \quad \Delta = \frac{V_1\dfrac{\partial V}{\partial r} - V\dfrac{\partial V_1}{\partial r}}{V_1\dfrac{\partial Z}{\partial r}}$$

eliminate both default risk and spot rate risk. The analysis results in

$$\frac{\partial V}{\partial t} + \tfrac{1}{2}w^2\frac{\partial^2 V}{\partial r^2} + (u - \lambda w)\frac{\partial V}{\partial r} - (r + \lambda_1(r,t)p)V = 0.$$

Observe that the 'market price of default risk' λ_1 is now where the probability of default appeared before, thus we have a risk-neutral probability of default. This equation is the risk-neutral version of Equation (23.4).

Can you imagine what happens if the risk of default is stochastic? There are actually three sources of randomness:

- the spot rate (the random movement in r)
- the probability of default (the random movement in p)
- the event of default (the Poisson process kicking in)

This means that we must hedge with *three* bonds, two other risky bonds and a risk-free bond, say. Where will you find market prices of risk? Can you derive a risk-neutral version of Equation (23.5)?

23.10 CREDIT RATING

There are many **credit rating agencies** who compile data on individual companies or countries and estimate the likelihood of default. The most famous of these are **Standard & Poor's** and **Moody's**. These agencies assign a **credit rating** or **grade** to firms as an estimate of their creditworthiness. Standard & Poor's rate businesses as one of AAA, AA, A, BBB, BB, B, CCC or Default. Moody's use Aaa, Aa, A, Baa, Ba, B, Caa, Ca, C. Both of these companies also have finer grades within each of these main categories. The Moody grades are described in the Table 23.2.

In Figure 23.5 is shown the percentage of defaults over the past 80 years, sorted according to their Moody's credit rating.

The credit rating agencies continually gather data on individual firms and will, depending on the information, grade/regrade a company according to well-specified criteria. A change of rating is called a **migration** and has an important effect on the price of bonds issued by the company. Migration to a higher rating will increase the value of a bond and decrease its yield, since it is seen as being less likely to default.

Clearly there are two stages to modeling risky bonds under the credit-rating scenario. First, we must model the migration of the company from one grade to another and second we must price bonds taking this migration into account.

Figure 23.6 shows the credit rating for Eastern European countries by rating agency.

Table 23.2 The meaning of Moody's ratings.

Aaa	Bonds of best quality. Smallest degree of risk. Interest payments protected by a large or stable margin.
Aa	High quality. Margin of protection lower than Aaa.
A	Many favorable investment attributes. Adequate security of principal and interest. May be susceptible to impairment in the future.
Baa	Neither highly protected nor poorly secured. Adequate security for the present. Lacking outstanding investment characteristics. Speculative features.
Ba	Speculative elements. Future not well assured.
B	Lack characteristics of a desirable investment.
Caa	Poor standing. May be in default or danger with respect to principal or interest.
Ca	High degree of speculation. Often in default.
C	Lowest-rated class. Extremely poor chance of ever attaining any real investment standing.

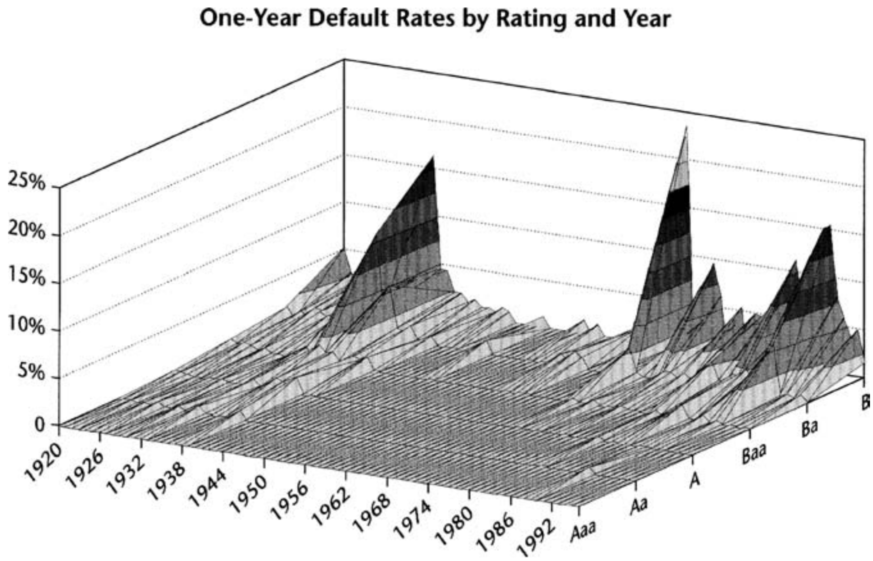
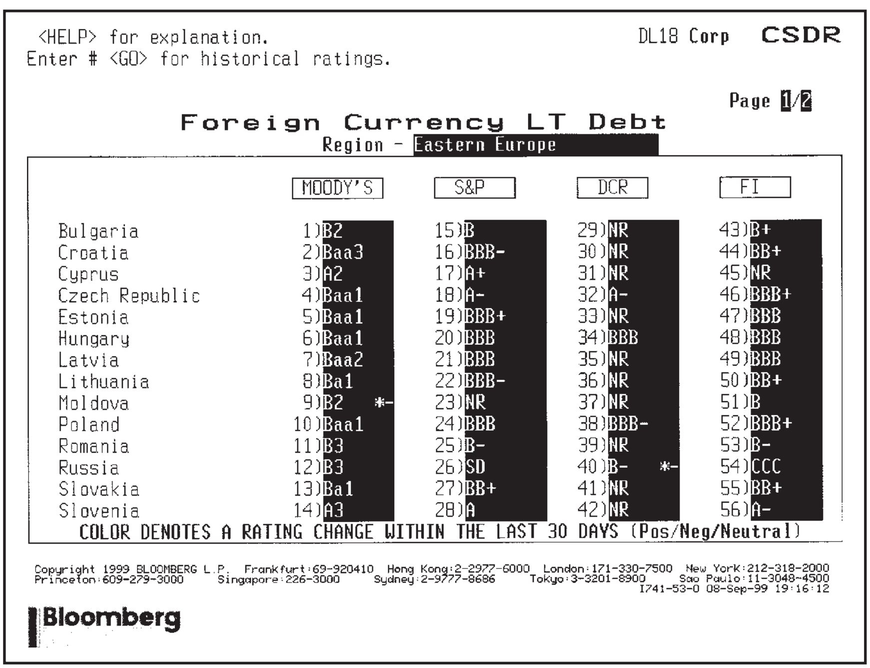


Figure 23.5 Percentage of defaults according to rating. Source: Moody's.



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Enter # <GO> for historical ratings.

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Foreign Currency LT Debt
Region - Eastern Europe

	MOODY'S	S&P	DCR	FI
Bulgaria	1) B2	15) B	29) NR	43) B+
Croatia	2) Baa3	16) BBB-	30) NR	44) BB+
Cyprus	3) A2	17) A+	31) NR	45) NR
Czech Republic	4) Baa1	18) A-	32) A-	46) BBB+
Estonia	5) Baa1	19) BBB+	33) NR	47) BBB
Hungary	6) Baa1	20) BBB	34) BBB	48) BBB
Latvia	7) Baa2	21) BBB	35) NR	49) BBB
Lithuania	8) Ba1	22) BBB-	36) NR	50) BB+
Moldova	9) B2 *-	23) NR	37) NR	51) B
Poland	10) Baa1	24) BBB	38) BBB-	52) BBB+
Romania	11) B3	25) B-	39) NR	53) B-
Russia	12) B3	26) SD	40) B- *-	54) CCC
Slovakia	13) Ba1	27) BB+	41) NR	55) BB+
Slovenia	14) A3	28) A	42) NR	56) A-

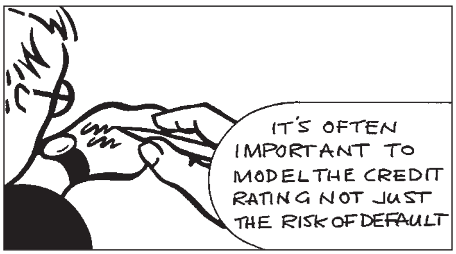
COLOR DENOTES A RATING CHANGE WITHIN THE LAST 30 DAYS (Pos/Neg/Neutral)

Bloomberg

Figure 23.6 Ratings for Eastern European countries by rating agency. Source: Bloomberg L.P.

Table 23.3 An example of a transition matrix.

	AAA	AA	A	BBB	BB	B	CCC	Default
AAA	0.90829	0.08272	0.00736	0.00065	0.00066	0.00014	0.00006	0.00012
AA	0.00665	0.90890	0.07692	0.00583	0.00064	0.00066	0.00029	0.00011
A	0.00092	0.02420	0.91305	0.05228	0.00678	0.00227	0.00009	0.00041
BBB	0.00042	0.00320	0.05878	0.87459	0.04964	0.01078	0.00110	0.00149
BB	0.00039	0.00126	0.00644	0.07710	0.81159	0.08397	0.00970	0.00955
B	0.00044	0.00211	0.00361	0.00718	0.07961	0.80767	0.04992	0.04946
CCC	0.00127	0.00122	0.00423	0.01195	0.02690	0.11711	0.64479	0.19253
Default	0	0	0	0	0	0	0	1



23.11 A MODEL FOR CHANGE OF CREDIT RATING

Company XYZ is currently rated A by Standard & Poor's. What is the probability that in one year's time it will still be rated A? Suppose that it is 91.305%. Now what is the probability that it will be rated AA or even AAA, or in default? We can represent these probabilities over the one year time horizon by a **transition matrix**. An example is shown in Table 23.3.

This table is read as follows. Today the company is rated A. The probability that in one year's time it will be at another rating can be seen by reading across the A row in the table. Thus the probability of being rated AAA is 0.092%, AA 2.42%, A 91.305%, etc. The highest probability is of no migration. By reading down the rows, this table can be interpreted as either a representation of the probabilities of migration of *all* companies from one grade to another, or of company XYZ had it started out at other than A. Whatever the grade today, the company must have some rating at the end of the year even if that rating is default. Therefore the probabilities reading across each row must sum to one. And once a company is in default, it cannot leave that state, therefore the bottom row must be all zeros except for the last number which represents the probability of going from default to default, i.e. 1.

This table or matrix represents probabilities over a finite horizon. But during that time a bond may have gone from A to BBB to BB, how can we model this sequence of migrations? This is done by introducing a transition matrix over an infinitesimal time period. We can model continuous-time transitions between states via **Markov chains**. This is quite an advanced topic, and I won't be going into this subject any more here.

23.12 COPULAS: PRICING CREDIT DERIVATIVES WITH MANY UNDERLYINGS

Although mentioned briefly above, credit derivatives with many underlyings have become very popular of late. As a natural skeptic when it comes to modeling in finance generally, I have to say that some of the instruments and models being used for the instruments fill me with some nervousness and concern for the future of the global financial markets. Not to mention mankind. Never mind, it's probably just me. We'll see the most important of these ideas and instruments now.

23.12.1 The copula function

The technique now most often used for pricing credit derivatives when there are many underlyings is that of the **copula**. The copula[2] function is a way of simplifying the default dependence structure between many underlyings in a relatively transparent manner. The clever trick is to separate the distribution for default for each individual name from the dependence structure between those names. So you can rather easily analyze names one at a time, for calibration purposes, for example, and then bring them all together in a multivariate distribution. Mathematically the copula way of representing the dependence (one marginal distribution per underlying, and a dependence structure) is no different from specifying a multivariate density function. But it can simplify the analysis.

The copula approach in effect allows us to readily go from a single-default world to a multiple-default world almost seamlessly. And by choosing the nature of the dependence, the copula function, we can explore models with richer 'correlations' than we have seen so far in the multivariate Gaussian world. For example, having a higher degree of dependence during big market moves is quite straightforward.

23.12.2 The mathematical definition

Take N uniformly distributed random variables $U_1, U_2, \ldots, U_N$, each defined on $[0, 1]$. The copula function (see Li, 2000) is defined as

$$C(u_1, u_2, \ldots, u_N) = \text{Prob}(U_1 \le u_1, U_2 \le u_2, \ldots, U_N \le u_N).$$

Clearly we have

$$C(u_1, u_2, \ldots, 0, \ldots, u_N) = 0,$$

and

$$C(1, 1, \ldots, u_i, \ldots, 1) = u_i.$$

So that's the copula function. The way it links many univariate distributions with a single multivariate distribution is as follows.

Let $x_1, x_2, \ldots, x_N$ be random variables with cumulative distribution functions (so-called **marginal** distributions) of $F_1(x_1), F_2(x_2), \ldots, F_N(x_N)$. Combine the Fs with the copula function,

$$C(F_1(x_1), F_2(x_2), \ldots, F_N(x_N)) = F(x_1, x_2, \ldots, x_N)$$

and it's easy to show that this function $F(x_1, x_2, \ldots, x_N)$ is the same as

$$\text{Prob}(X_1 \le x_1, X_2 \le x_2, \ldots, X_N \le x_N).$$

A key reference in this subject is Sklar (1959) who showed that any multivariate distribution can be written in terms of a copula function, i.e. the converse of the above.

In pricing basket credit derivatives we would use the copula approach by simulating default times of each of the constituent names in the basket. And then perform many such simulations in order to be able to analyze the statistics, the mean, standard deviation, distribution, etc., of the present value of resulting cashflows.

[2] From the Latin for 'join.'

23.12.3 Examples of copulas

Here are some examples of bivariate copula functions. They are readily extended to the multivariate case.

Bivariate Normal

$$C(u,v) = N_2\left(N_1^{-1}(u), N_1^{-1}(v), \rho\right), \quad -1 \le \rho \le 1,$$

where N_2 is the bivariate Normal cumulative distribution function, and N_1^{-1} is the inverse of the univariate Normal cumulative distribution function.

Frank

$$C(u,v) = \frac{1}{\alpha}\log\left(1 + \frac{(e^{\alpha u}-1)(e^{\alpha v}-1)}{e^{\alpha}-1}\right), \quad -\infty < \alpha < \infty.$$

Fréchet–Hoeffding upper bound

$$C(u,v) = \min(u,v).$$

Gumbel–Hougaard

$$C(u,v) = \exp\left(-\left((-\log u)^{\theta} + (-\log v)^{\theta}\right)^{1/\theta}\right), \quad 1 \le \theta < \infty.$$

This copula is good for representing extreme value distributions.

Product

$$C(u,v) = uv$$

Also: Archimedean, Clayton, Student. See Cherubini *et al*. (2004) for a comprehensive list.

One of the simple properties to examine with each of these copulas, and which may help you decide which is best for your purposes, is the **tail index**. Examine

$$\lambda(u) = \frac{C(u,u)}{u}.$$

This is the probability that an event with probability less than u occurs in the first variable given that at the same time an event with probability less than u occurs in the second variable. Now look at the limit of this as $u \to 0$,

$$\lambda_L = \lim_{u\to 0}\frac{C(u,u)}{u}.$$

This tail index tells us about the probability of both extreme events happening together.

23.13 COLLATERALIZED DEBT OBLIGATIONS

Whereas most of the credit instruments described above have a single underlying name, the **collateralized debt obligation** or **CDO** is an instrument designed to give protection against losses in a portfolio, and usually a portfolio containing hundreds of individual companies. As with other credit derivatives there is the protection buyer and the seller.

The CDO gives protection, not against an entire portfolio, but against bits of it, so-called **tranches**. First, define the aggregate loss in the portfolio as the sum of all losses due to default. As more and more companies default so the aggregate loss will increase. Now specify hurdles, as percentages of notional, and these define the tranches. For example, there may be the 0–3% tranche and the 3–7% tranche, etc. As the aggregate loss increases past each of the 3%, 7%, etc. hurdles so the owner of that tranche will begin to receive compensation, at the same rate as the losses are piling up. You will only be compensated once your **attachment point** has been reached and until the **detachment point**.

So there are two types of cashflows in this instrument:

- Premiums: The protection buyer will pay periodic premiums to the seller, quoted in basis points. There may also be an upfront premium.
- 'Compensation:' The holder of the instrument, the protection buyer, receives the loss suffered by his tranche.

To clarify, losses are assigned initially to the first tranche until the threshold is attained, and then to the second tranche until its detachment point is reached, and so on. An example is given in Table 23.4.

To price these instruments you obviously have to consider the probability of each name defaulting and the correlation structure between them. A model often used in practice is the Gaussian copula approach. So far so good. To make the problem tractable, given the large number of underlyings and parameters, it is also often assumed that the structure of the correlation is described by a single random factor. Each has its own stock-specific source of randomness but there is a factor common to all names. Then, to simplify matters further, it is also often assumed that there is a single number to represent all of these correlations, that is, there is just one single correlation parameter. Probably (at least) one simplification too far. One of the reasons for such assumptions is that it leads to simple pricing formulæ.

Working within the framework of a single parameter, how is this parameter used? Suppose we want to price a new CDO, and we have the price of a traded CDO that is not dissimilar. We can back out the **implied** or **compound correlation** from the CDO with the known price and plug it into the model for the other CDO. This is, of course, what is often done in the derivatives world, equity, FX or fixed income, with volatility. It's just another form of calibration.

Table 23.4 A typical CDO and its tranches. Source: St Pierre (2004).

Tranche	Upfront premium	Ongoing premium (bps)
0–3%	42%	500
3–7%	0%	331
7–10%	0%	126
10-15%	0%	54
15–30%	0%	16

There is a major problem with this. As happens in the derivatives world, non-credit, we find that we need a different implied correlation for each tranche. Typically, you might find that the implied correlation falls from first tranche to second, and then rises for third and subsequent tranches (Finger, 2004). This is akin to the implied volatility smile and skews that we have seen before. This suggests that the market is somehow allowing for things not within the one-size-fits-all correlation assumption. Slightly concerning, but not too much. No, the real worry is that you may find that there are *two* possible implied correlations, or no implied correlation at all. This means that a market price is either consistent with two possible constant correlations, or there is no constant correlation which matches theoretical and market prices. Generally speaking, if we were to plot value of a tranche versus implied correlation then we might find a non-monotonic curve. Why is this a worry? Think back to examples we've seen with implied volatility. In particular, the up-and-out call option that we analyzed in some detail (see Chapters 8 and 13). We had the same problem there. Before I expand on this in one paragraph's time, let's see one final way of pricing CDOs.

Using the example in the table, you can think of the 3–7% tranche as long a 0–7% tranche and short the 0–3% tranche. So you can value a non-existent 0–7% tranche very easily. Back out from market data for the 0–3% tranche the implied correlation for that first tranche. Now back out an implied correlation for the 0–7% tranche that you have created. This is called the **base correlation**. Now use these base correlations to price other CDOs with tranches 0–x%, and then construct the y%–x% tranches accordingly.

Now back to the problem with all of this. Do you remember what the non-uniqueness in the implied volatility for the up-and-out call meant? It was because the price versus (constant) volatility graph was non-monotonic. It rose to a peak and then fell. If we have that behavior with a CDO then alarms bells should start ringing. In the knockout example the non-monotonicity was because there were two competing effects going on. If we were out of the money we wanted high volatility so we moved into the money and hence got a nice payoff. But once in the money we wanted low volatility so that we didn't knock out and lose the option. So sometimes volatility was good for us and sometimes bad. Now think bigger than barrier options: two competing effects, one good, one bad. There must be something similar going on with the CDO. Sometimes correlation is good, sometimes bad. What if the good thing didn't kick in and we were only left with the bad? That's what we looked at when we examined the barrier option, we said what if the good volatility, out of the money, didn't appear, but volatility only rose in the money where it hurt us. Net result, much, much lower barrier option prices than if volatility were to rise uniformly everywhere. Exactly the same thing is happening with CDOs. What if the good correlation didn't appear, what if only the bad correlation rose? Ouch. Prices could be much, much lower than you would expect.

With so many potential variables and parameters because of the hundreds of underlyings it is going to be much harder to examine more interesting scenarios. But a step in the right direction would be to at least have two sources of randomness leading to correlations between the underlying names. Or to vary correlation with burn-through (aggregate loss).

23.14 SUMMARY

As can be seen from this chapter, credit risk modeling is a very big subject. I have shown some of the popular approaches, but they are by no means the only possibilities. To aid with the assessment of credit risk, the bank JP Morgan has created CreditMetrics, a methodology for assessing the impact of default events on a portfolio. This is described in Chapter 24.

As a final thought for this chapter, suppose that a company issues just the one risky bond so that there is no way of hedging the default. If you believe that the market is underpricing the bond because it overestimates the risk of default then you might decide to buy it. If you intend holding it until expiry, then the market price in the future is only relevant in so far as you may change your mind. But you really do care about the likelihood of default and will pay very close attention to news about the company. On the other hand, if you buy the bond with the intention of only holding it for a short time your main concern should be for how the market is behaving and the real risk of default is irrelevant. You may still watch out for news about the company, but now your concern will be for how the market reacts to the news, not the news itself.

FURTHER READING

- See Black & Scholes (1973), Merton (1974), Black & Cox (1976), Geske (1977) and Chance (1990) for a treatment of the debt of a firm as an option on the assets of that firm. See Longstaff & Schwartz (1994) for more recent work in this area.
- The articles by Cooper & Martin (1996) and Schönbucher (1998) are general reviews of the state of credit risk modeling.
- See Apabhai *et al*. (1998) for more details of the company and debt valuation model, especially for final and boundary conditions for various business strategies. Epstein *et al*. (1997 a, b) describe the firm valuation models in detail, including the effects of advertising and market research.
- The classic reference, and a very good read, for firm-valuation modeling is Dixit & Pindyck (1994). For a non-technical POV see Copeland *et al*. (1990).
- See Kim (1995) for the application of the company valuation model to the question of company mergers and some suggestions for how it can be applied to problems in company relocation and tax status.
- Important work on the instantaneous risk of default model is by Jarrow & Turnbull (1990, 1995), Litterman & Iben (1991), Madan & Unal (1994), Lando (1994a), Duffie & Singleton (1994 a, b) and Schönbucher (1996).
- See Blauer & Wilmott (1998) for the instantaneous risk of default model and an application to Latin American Brady bonds.
- See Duffee (1995) for other work on the estimation of the instantaneous risk of default in practice.
- The original work on change of credit rating was due to Lando (1994b), Jarrow *et al*. (1997) and Das & Tufano (1994). Cox & Miller (1965) describe Markov chains in a very accessible manner.
- See Ahn *et al*. (1998) for the rather sensible use of utility theory in credit risk modeling.

- Current market conditions and prices for Brady bonds can be found at `www.bradynet.com`.
- See `www.emgmkts.com` for financial news from emerging markets.
- Clark (2002) uses the implied volatility as a measure of country risk.

EXERCISES

1. A risk-free government, zero-coupon bond, with a principal of \$100, maturing in two years has a value of \$91.75. A risky corporate zero-coupon bond with the same principal and maturity is worth \$88.25. What is the market's estimate of the risk of default assuming zero recovery?
2. The same government and company as in the above question now issue five-year bonds with \$100 principal and prices for government bond and corporate bond of \$82.15 and \$78.89 respectively. What does this now say about the market's view on the probability of the company defaulting?
3. Repeat the analyses of the above two problems but with the extra assumption that on default their is a recovery rate of 10%, i.e. you will receive \$10 at maturity if there has been default.
4. Construct the intermediate steps in the derivation of the equation for the value of a risky bond (when default is governed by a Poisson process):

$$\frac{\partial V}{\partial t} + \frac{1}{2}w^2\frac{\partial^2 V}{\partial r^2} + (u - \lambda w)\frac{\partial V}{\partial r} - (r + p)V = 0.$$